

CloudKitchen Flow

AI-Ready Cloud Kitchen Ordering & Kitchen Automation Platform

Project Report + SRS + Overleaf Source

Prepared for Project Development

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1 Abstract

CloudKitchen Flow is a workflow-driven cloud kitchen platform that allows customers to browse a menu, place orders, track order status, and receive delivery updates, while enabling kitchen staff and administrators to manage the full order lifecycle from a unified dashboard. The system is designed as a practical automation project using a free-stack architecture built around React for the frontend, n8n for workflow automation, and Google Sheets or Supabase as the backend datastore for MVP development. The platform supports three roles: customer, kitchen staff, and admin.

The project focuses on automating the full operational journey of a cloud kitchen: menu retrieval, order placement, kitchen acceptance, preparation updates, dispatch, and delivery completion. In addition to the core transactional flow, the platform is designed to be AI-ready, allowing future extensions such as smart menu recommendations, ETA prediction, automated support replies, and daily business analytics.

2 Introduction

2.1 Background

Cloud kitchens are delivery-first food businesses that operate without traditional dine-in infrastructure. Their operations depend heavily on efficient order intake, kitchen coordination, and delivery status tracking. Small and medium cloud kitchens often rely on fragmented tools such as messaging apps, spreadsheets, and manual phone coordination, leading to delays, missed orders, poor visibility, and weak customer experience.

2.2 Problem Statement

A typical small cloud kitchen lacks a centralized, affordable order management platform that can:

- collect customer orders digitally,
- store and track order data in real time,
- provide kitchen staff with a simple action dashboard,
- update order statuses consistently,
- allow customers to track their order journey,
- provide admins with visibility into sales, pending orders, and menu management.

2.3 Proposed Solution

CloudKitchen Flow solves this by providing:

- a customer-facing ordering interface,
- a kitchen dashboard for operational order handling,
- an admin dashboard for menu and analytics,

- n8n workflow automation for all backend business logic,
- a simple free-stack deployment model suitable for student projects and MVPs.

3 Objectives

- Build a complete cloud kitchen ordering platform with customer, kitchen, and admin interfaces.
- Automate the order lifecycle using n8n workflows.
- Store and retrieve menu and order data using Google Sheets or a similar free backend.
- Enable live order status tracking for customers.
- Provide dashboards for kitchen staff and administrators.
- Design the system so that AI-powered features can be added in future versions.

4 Scope

4.1 In Scope

- Menu listing and browsing
- Cart and checkout
- Order placement and order ID generation
- Order tracking by order ID
- Kitchen acceptance/rejection of orders
- Kitchen status updates: Preparing, Ready
- Delivery status updates: Out for Delivery, Delivered
- Admin menu management
- Admin order overview and basic analytics

4.2 Out of Scope for MVP

- Multi-kitchen marketplace support
- Real-time delivery rider location tracking
- Full payment gateway integration
- Native mobile app
- Enterprise-grade inventory management

5 System Overview

5.1 User Roles

1. Customer

- View menu
- Add items to cart
- Place orders
- Track orders

2. Kitchen Staff

- View new orders
- Accept or reject orders
- Update preparation and dispatch statuses
- Mark delivery completion

3. Admin

- Manage menu items
- View all orders
- Monitor sales and operational metrics

5.2 High-Level Workflow

1. Customer opens menu and selects items.
2. Customer places order through checkout form.
3. n8n receives order via webhook and stores it in Orders sheet.
4. Kitchen dashboard fetches pending orders.
5. Kitchen accepts/rejects order and updates status.
6. Customer can track current order state.
7. Admin monitors performance and manages menu items.

6 Technology Stack

Layer	Technology	Purpose
Frontend	React + Vite + Tailwind CSS	Customer UI, Kitchen dashboard, Admin dashboard
Workflow Backend	n8n	Webhooks, business logic, workflow automation
Database (MVP)	Google Sheets	Menu and order storage

Alternative Database	Supabase	Better scalability for later versions
API Communication	Axios / Fetch	Frontend to n8n webhook communication
Hosting	Vercel / Render / Railway / local n8n	Deployment of frontend and workflow engine
Notifications (optional)	Gmail / Twilio / WhatsApp API	Order notifications and customer updates

7 Module Description

7.1 Customer Module

- Menu browsing page
- Search and category filtering
- Cart management
- Checkout form
- Order success screen
- Track order page

7.2 Kitchen Module

- Dashboard with order counts
- New Orders page
- Active Orders page
- Completed Orders page
- Action buttons for status changes

7.3 Admin Module

- Dashboard with order analytics
- Menu management
- Orders overview
- Add/edit menu item support

8 Database Design

8.1 Menu Sheet

Field	Type	Description
item_id	String	Unique menu item ID
item_name	String	Name of food item
category	String	Category such as Pizza, Burger, Rolls
price	Number	Price of item
available	String	Yes/No availability
prep_time	Number	Estimated preparation time in minutes
image_url	String	Optional image link
created_at	DateTime	Creation timestamp

8.2 Orders Sheet

Field	Type	Description
order_id	String	Unique order ID
customer_name	String	Customer name
phone	String	Contact number
address	String	Delivery address
items	JSON/String	Ordered items list
total_amount	Number	Total bill amount
payment_mode	String	COD / future online mode
order_status	String	Placed / Accepted / Rejected / Completed
kitchen_status	String	Order Received / Preparing / Ready
delivery_status	String	Pending / Out for Delivery / Delivered
created_at	DateTime	Order creation timestamp
updated_at	DateTime	Last update timestamp
accepted_at	DateTime	Optional accepted time
ready_at	DateTime	Optional ready time
delivered_at	DateTime	Optional delivery time

9 n8n Workflow Design

9.1 Workflow 1: Get Menu

Endpoint: GET /menu

Steps:

1. Webhook trigger receives GET request.
2. Google Sheets node reads Menu sheet.
3. Set node formats response.
4. Respond to Webhook returns menu JSON.

9.2 Workflow 2: Place Order

Endpoint: POST /place-order

Steps:

1. Webhook receives checkout payload.
2. Set/Code node generates order ID and total amount.
3. Default statuses are assigned.
4. Google Sheets appends a new row in Orders sheet.
5. Response returns order confirmation and order ID.

9.3 Workflow 3: Track Order

Endpoint: GET /track-order?order_id=...

Steps:

1. Webhook receives order ID.
2. Google Sheets searches order row.
3. Set node formats order detail response.
4. Respond to Webhook returns status data.

9.4 Workflow 4: Get Kitchen Orders

Endpoint: GET /kitchen-orders

Steps:

1. Webhook receives request.
2. Google Sheets reads all orders.
3. Set node prepares data for dashboard.
4. Response returns kitchen order list.

9.5 Workflow 5: Update Order Status

Endpoint: POST /update-order-status

Accepted actions:

- accept
- reject
- preparing
- ready
- out_for_delivery

- delivered

Action Mapping:

- accept $\rightarrow$ order_status = Accepted
- reject $\rightarrow$ order_status = Rejected
- preparing $\rightarrow$ kitchen_status = Preparing
- ready $\rightarrow$ kitchen_status = Ready
- out_for_delivery $\rightarrow$ delivery_status = Out for Delivery
- delivered $\rightarrow$ delivery_status = Delivered, order_status = Completed

9.6 Workflow 6: Add Menu Item

Endpoint: POST /add-menu-item

Adds a new menu item to Menu sheet.

9.7 Workflow 7: Update Menu Item

Endpoint: POST /update-menu-item

Looks up item by item_id and updates relevant fields.

10 Frontend Pages**10.1 Customer Pages**

- Menu Page
- Cart Page
- Checkout Page
- Order Success Page
- Track Order Page

10.2 Kitchen Pages

- Kitchen Dashboard
- New Orders Page
- Active Orders Page
- Completed Orders Page

10.3 Admin Pages

- Admin Dashboard
- Menu Management Page
- Orders Overview Page

11 Software Requirements Specification (SRS)

11.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements for CloudKitchen Flow, a cloud kitchen ordering and order-management platform.

11.2 Intended Audience

- Project mentors and evaluators
- Developers implementing the frontend and workflows
- Stakeholders reviewing scope and features

11.3 Product Perspective

CloudKitchen Flow is a standalone web-based platform with automation workflows. It integrates a frontend application with n8n webhook-based business logic and a spreadsheet-backed data store for MVP use.

11.4 Functional Requirements

11.4.1 Customer Requirements

FR-C1 The system shall display available menu items to customers.

FR-C2 The system shall allow customers to add items to cart and modify quantities.

FR-C3 The system shall allow customers to submit an order with contact and address details.

FR-C4 The system shall generate a unique order ID for every order.

FR-C5 The system shall allow customers to track orders using order ID.

11.4.2 Kitchen Requirements

FR-K1 The system shall display newly placed orders to kitchen staff.

FR-K2 The system shall allow kitchen staff to accept or reject orders.

FR-K3 The system shall allow kitchen staff to update order preparation status.

FR-K4 The system shall allow kitchen staff to mark an order out for delivery and delivered.

11.4.3 Admin Requirements

FR-A1 The system shall allow administrators to view all orders.

FR-A2 The system shall allow administrators to add and update menu items.

FR-A3 The system shall display operational analytics such as total orders and revenue.

11.5 Non-Functional Requirements

NFR-1 The system should be responsive across desktop and mobile screens.

NFR-2 The system should provide a simple and intuitive user interface.

NFR-3 The system should respond to common actions within acceptable time for MVP usage.

NFR-4 The architecture should support future migration from Google Sheets to a proper database.

NFR-5 The workflow design should be modular and easy to extend with AI agents.

11.6 Assumptions and Dependencies

- n8n instance is available and accessible to the frontend.
- Google Sheets API credentials are configured correctly.
- Frontend environment variables are configured with webhook base URL.
- Kitchen and admin users are trusted internal users in MVP stage.

12 Use Case Summary

Actor	Use Case	Description
Customer	Browse Menu	View all available menu items
Customer	Place Order	Submit order from cart via checkout
Customer	Track Order	Check order status using order ID
Kitchen Staff	Accept/Reject Order	Process newly received orders
Kitchen Staff	Update Status	Mark preparing, ready, out for delivery, delivered
Admin	Manage Menu	Add or update food items
Admin	Monitor Operations	View dashboards, orders, and analytics

13 Future Enhancements

- AI menu recommendation assistant
- ETA prediction based on kitchen load
- Automated order support chatbot

- Inventory alerts and stock management
- Payment gateway integration
- Multi-kitchen support
- Delivery rider assignment panel

14 Conclusion

CloudKitchen Flow is a strong workflow automation project that demonstrates how a real-world food ordering business can be digitized using a low-cost and scalable architecture. It combines frontend application design, backend workflow automation, structured data storage, and future AI extensibility in a single project. The project is well-suited for academic demonstration, MVP experimentation, and portfolio use.